Felipe REIS

Paris, France • felipe.reis-campanha-ribeiro@polytechnique.edu • +33 07 69 74 01 56 • linkedin.com/in/felipercr/

Master student specialized in **signal processing, AI, and embedded/critical systems**. Looking for challenging master thesis or R&D internships in industries such as aerospace, defense, and communications, around April to August**. Fluent in English, French** Portuguese and Spanish.

EDUCATION

École Polytechnique – Cyber-physical systems | *Master's degree (France)* — *Sep 2025 – Aug 2026*

Télécom SudParis – Computer science | *Cycle ingénieur (France)* — *Sep 2024 – Aug 2026*

Federal University of Minas Gerais – Electrical engineering | *Bachelor's degree, 5-years (Brazil)* — *Jan 2021 – Dec 2026*

University of Jyväskylä – Data science and AI | *Exchange period (Finland)* — *Jan 2024 – May 2024*

PROFESSIONAL EXPERIENCE

Research Internship – *Electronics and Physics Department at Télécom SudParis (France)* — *Jan 2025 – Aug 2025*

- Study of novel signal processing techniques for active sonars inspired by dolphin echolocation.
- Implementation in MATLAB of **AI, signal processing, and optimization algorithms**. Perform acoustic simulations.
- **Work presented to the DGA.** Writing an **original article** to be published in a scientific journal.

Research Internship – *Spectral Imaging Laboratory at the University of Jyväskylä (Finland)* — *Feb 2024 – May 2024*

- Characterization of a hyperspectral imaging camera for applications in **space**.
- Evaluation of the camera's technology through an **analysis of a large dataset of hyperspectral images** using Python.
- Presented in **IEEE Whispers 2024**. Co-writer of scientific article [1].

Peer Tutor, Signals & Systems – *Electronics Department, UFMG (Brazil)* — *Feb 2023 – Dec 2023*

- Aid students in topics such as Fourier analysis, filters, signal processing, control systems, differential equations, etc.

R&D Intern – *Metrum, Measuring and Testing Equipment Ltd. (Brazil)* — *May 2023 – Nov 2023*

- **Develop** equipment for measuring and testing electronics for use in electrical, communication and automation companies.
- Implement algorithms on **DSPs and microcontrollers**. Design and implement **electronic circuits**.
- Won bidding processes from energy and telecommunication companies.

Programmer and Electrical Engineer – *Milhagem UFMG (Brazil)* — *May 2023 – Nov 2023*

- Competition team that makes energy-efficient vehicles to participate in the Shell-Eco Marathon.
- Work on the vehicle's **telemetry system and motor driver** using C/C++. Programed a **remote measuring system** using 5G. Developed **time-critical** embedded programming for motor driver.
- **1st place** in the Shell-Eco Marathon Americas 2023.

Research Internship – *Nuclear Technology Development Center CDTN (Brazil)* — *Oct 2022 – Oct 2023*

- Validation of a method for calculating error of stochastic neutronic simulations.
- **Data analysis** of simulations from a nuclear reactor using Python (Matplotlib, Jax, NumPy, etc.).
- Co-writer of scientific article [2].

SKILLS

Programming: Python, C/C++, MATLAB, Java, Git, Linux, NVIDIA Cuda, Docker, SQL.

Data Science & AI: Python and MATLAB packages (JAX, Keras, TensorFlow, Matplotlib, etc.), optimization solvers, OpenCV, neural networks, AI, machine learning, computer vision, image analysis, numerical methods, compression.

Electronics: DSPs, microcontrollers, filters, analog/digital electronics, FPGAs, Verilog, VHDL, radio systems, control systems.

Mathematics: Fourier, Laplace, Z, DCT, wavelets, optimization, differential equations, compressed sensing, probability.

LANGUAGES

Fluent in English, French, Portuguese, and Spanish

CENTER OF INTEREST

I love music and have been playing guitar as a hobby for 12 years, 6 in a band. I also love to travel and to experience different cultures.

PUBLICATIONS

[1] L. Lind, F. R. C. Ribeiro, H. Saari, *et al*., "Lifetime Performance Analysis of a Hyperspectral Imager Based on a Piezoelectrically Tunable Fabry-Perot Interferometer," *Sensing and Imaging*, vol. 26, p. 105, 2025, doi: 10.1007/s11220-025-00628-z.

[2] T. A. S. Vieira, F. R. C. Ribeiro, Y. M. Carvalho, V. V. A. Silva, G. De Paula Barros, and A. A. C. D. Santos, "Investigation of discretization uncertainty in Monte Carlo neutron transport simulations of the Molten Salt Fast Reactor (MSFR)," *Brazilian Journal of Radiation Sciences*, vol. 11, no. 4, pp. 1–27, Dec. 2023, doi: 10.15392/2319-0612.2023.1317.